

Human Impact on the Climate and Sustainability

Year after year, the climate deteriorates. Global warming ravages the environment, and we feel impotent as it happens. We see, read and hear of it everywhere, and we feel that big corporations are to blame, so what can we do? If it is not us to blame, how can we backtrack the actions and lessen the impact?

The first step would be to accept that we *are* to blame.

Statistically, you could be the reason for a large chunk of the climate change. According to the Carbon Majors Database, about 10-12% of the sources contributing to climate change comes from habits in our day-to-day lives that we practice and could easily get rid of. 10% does not seem like a lot. But the truth is, 10% is actually a huge number; it equals 3,800,000,000 metric tons of carbon dioxide in the atmosphere every year.

Furthermore, by removing that 10-12%, it causes the manufacturing processes to reduce their output by 4%. What does that mean? Why manufacturing? Why is that number important? That number is important because manufacturing processes are doing it to produce the things we want. If we reduce our consumption of the world's resources that we do not need, the manufacturing plants would reduce their emissions because the demand is less. If the entire world reduced their consumption rate by even 10%, it would cause a 4% decrease in emissions from the entire global manufacturing sector. 4% of the world's emissions is an outrageous number. So, if we have the ability to reduce emissions, where do we start?

Let's look at what exactly we are doing wrong, and how we could improve our habits:

Let's start with the most obvious offender: transportation. One of the easiest ways to help the climate is to walk or cycle to places.

Think about it: how many little trips in your car under a mile do you take every week? How often do you get in your car just to drive to the grocery store to pick up a snack you didn't have at home? How many times have you gotten in the car just so you wouldn't have to walk outside to get the mail? It's much healthier for you to walk short distances and it does wonders for the environment.

Let's look at the numbers, because this turns out to be a horrible trade-off. For a car to be effective, the engine has to be at a certain temperature before it can properly burn the fuel that keeps it running. For short-distance driving, the car will run at

lower temperatures throughout the trip, wasting fuel and putting out more emissions. To offset the bad impact on the environment of driving your car on a short little trip, you would have to buy 24 items at the store to make the carbon footprint of that trip worthwhile. Driving a car to pick up a bag of chips or an energy drink is a terrible deal for your environment. If you're going less than a mile, just walk.

Let's talk about the fridge. We have this horrible habit in our society of going to the store, buying everything we could possibly need for the next month, and then forgetting about it. In global measurements, we waste about a third of the food produced in the world. Think about that for a second. Think about going to the store and buying three bags of groceries, then throwing one of them away before you get to your car. That is what we do, globally. The bad thing is the food normally gets tossed into a landfill, where it slowly gets compressed by surrounding junk. This particular type of decomposition produces methane, which is an especially potent greenhouse gas. If you think carbon dioxide is bad, you are yet to hear the effects of methane. In terms of impact on the climate, methane is approximately 28 times worse than carbon dioxide.

And the solution is easy: do a check in on the food in your fridge and move things to the front that you might want to eat soon. Get rid of the food that has been forgotten and will likely just rot in the back of the fridge. By doing this you would be preventing a large quantity of food from going to landfills and releasing greenhouse gases. That's healthier for the earth, and healthier for your wallet.

The production of beef is incredibly harmful to the environment. It takes an obscene amount of resources to make beef, compared to other meats and certainly compared to plants. The production of one kilogram of beef results in the creation of 60 kilograms of greenhouse gases and uses thousands of liters of water. Cattle require much larger amounts of land to be cleared so that they have something to graze on, and cows create massive amounts of methane to add on top of that. The premise is not to boycott beef, just try to limit your consumption. If you cut out beef on two or three days out of the week, you will have personally saved hundreds of thousands of liters of water and reduced your carbon footprint by a shocking amount. That is basically the equivalent to taking your car off the road for a few months. It is an amazing impact in return to a small change in your diet.

Your TV, computer, microwave clock, wireless router, all of these devices have something in common. You think that they are off, but they are actually still sucking

electricity from an outlet 24/7. This is known to scientists as “Vampire Energy” or energy that is used from electronic devices that are switched off but are still plugged in. It all adds up to approximately 5-10% of the electricity used in the average home. Where does this electricity come from? In most countries, it is supplied in from a local power grid. Most of the electricity coming from these power grids is generated by either coal or natural gas. All you need to do is buy a smart power strip to plug your electronics into, or unplug your phone chargers when your phone reaches a 100%.

It seems as though we have gotten incredibly spoiled with the technology we have at our fingertips. If it is cold, we blast the heater. If hot, we blast the cooler. We waste so much energy trying to heat buildings and cool buildings down to an artificial temperature. Heating and air conditioning are the largest contributors to the amount of energy used in an average household. If you just reduced the temperature of your home by one or two degrees depending on the season, you would reduce the overall load of your heating and air conditioning systems by 10%. Pair that with washing your clothes in cold water (This reduces the overall energy needed to wash a single load by 90%!) and you’ve completely optimized your home’s energy load without making any major lifestyle changes.

“The Cloud” is not a magic grid of servers that makes uses solar panels and wind turbines that power up every time someone streams a movie. The reality is that these warehouses full of servers run 24/7 and use electricity to power servers that keep our websites up and running. When you leave an application running on your computer or your phone, or you leave a browser window open just because you were going to look at it later, you’re forcing these warehouses to run needlessly and expend extra energy just for you. Another bad habit is leaving old, unused emails in your inbox and never deleting them. Emails and websites that are hosted on “The Cloud” are especially bad for the environment.

Simply put a cap on your streaming resolution. If you watch a movie on your phone or computer in 4k knowing that you will not pay it much attention, it is much better for the environment to just watch it in 1080p. Your eyes won’t even notice the difference!

Is your wardrobe bursting at the seams with clothes? In the last decade, the fashion industry has shifted from releasing new clothes four times a year to releasing dozens of smaller collections every week. These stores churn out over 100 billion pieces of clothing every year. In order to make these clothes for such a small

amount of time, fast fashion stores use low quality, synthetic fibers like polyester and nylon to make sure that the clothes can last for a while. Did you know that polyester is made from oil and natural gas? Cheap shirts, worn twice, tossed away, end up in a landfill and take hundreds of years to decompose. It has been found that washing a polyester item to release hundreds of thousands of micro plastics into the environment. If your favorite shirt gets a hole in it, instead of throwing it away, look up a tutorial and patch it up. By limiting consumption, we can reduce the frantic production schedules of fast fashion retailers and cut emissions from textile mills and transport at the same time.

We believe that as long as we toss all of our plastic water bottles and iced coffee cups in the recycling bin, everything will be fine. The truth is, there is only a 9% chance that any given piece of plastic is actually going to be recycled. The other 91% of plastic waste in the world is either buried in landfills or burned in the atmosphere or in the oceans to make way for new plastic waste.

You see, plastic does not come from anywhere magical. It comes directly from petrochemicals. Every time you use a single-use plastic bag for 10 minutes and then toss it away, you are fueling the machine of global plastic production and encouraging more and more plastic to be produced.

You do not have to change your entire life habits in one day, but it is important that we start building anchor habits. Buy a nice, reusable water bottle and a few reusable grocery bags today and put them in your bag or in your car so that you remember them and use them.

Companies work tirelessly to figure out ways to make things easier on the consumer so that you don't have to think about something hard like "Do I really need this?" before you buy it. One click on a mouse and the next day a truck will drive across town and leave a tiny plastic-wrapped package at your door. It's absolutely ridiculous how much we've fallen for this marketing.

In every decision that we make, we have to be vigilant. Whenever we are presented with the option to be lazy and consume more, we have to remember that we are voting for a dirtier planet. Before we buy that little trinket we saw online from a huge-box corporation, we have to ask ourselves if walking to the corner store and buying it ourselves would have been a reasonable substitute. Would it really have been worth it to have it shipped to us in a box of plastic? Corporations don't care about anything besides their stock prices, so if we remove a few percentage points from their market share by reducing consumption, they will be the first ones to adapt their

marketing plan to fit the new world order. You have much more power than you think.

We've all heard that we shouldn't leave the water running while we brush our teeth. We've all heard that it's bad for the environment to waste water, but what a lot of people don't realize is that it's not just the water that's wasteful. The amount of energy it takes to move and process water is even more astounding.

Every single drop of water you use in your home has to go through a massive filtration process at a municipal water plant and then be pumped through underground water pipes into your home. If you use hot water, your home has to use gas or electricity to heat it and make it ready to use.

That's right, all the water in your home has to be moved to you and treated in some way before it can be used. Leaving the water running unnecessarily or taking a twenty minute long hot shower is going to increase the amount of energy your house has to use to give you water. By just reducing your shower time by two minutes and putting flow restrictors on your sink, you will reduce the amount of energy needed to provide you with water without having to change your lifestyle at all. Reducing unnecessary water consumption is a great way to reduce your carbon footprint because you're lowering the overall load of energy put out by your local power plants.

We've conditioned ourselves to believe that things should be thrown away after being used only once. Paper towels, plastic forks and knives, cheap plastic storage containers, cardboard sleeves for coffee cups, they're all meant to be tossed away after only being used for a short time.

Think about it. A paper towel is used to clean up a water stain on your countertop, and then it's tossed away. Before it got to your hands, it had to be cut down from a tree, shipped to a factory, treated with chemicals, wrapped in plastic, shipped to the store, and driven home by you. That's an absolutely ridiculous amount of energy to put into something that's only supposed to live for a few seconds.

This is the reality of our disposable culture. This is why we need to start switching to reusable items. Instead of using paper towels, use cloths. Instead of cheap plastic storage containers, use cloth bags. When we stop demanding disposable items, we take away the fuel for a large chunk of the global manufacturing industry and reduce the amount of greenhouse gases being put into the atmosphere.

The worst thing you could do for the climate right now is to fall into the "Climate Doom" mentality. It's a very real thing that we all fall victim to. We see the climate

crisis, and we think that it's much too late to fix anything and that if the world is ending anyway, maybe we should just enjoy ourselves while we can. That's a very real viewpoint that a lot of people see, fall into, and accept as a lifestyle. This is a terrible way to think, because it makes us much less likely to make the changes that will save us. Instead, think positively. Nobody is expecting you to be more than human. All the world is asking for is millions and millions of people all over the globe to make minuscule changes to their everyday lives and accept that the changes they have to make aren't going to be easy.

Switching from a corded phone to a cordless phone, bunching together your errands into one trip rather than spreading them out, walking to the coffee shop to get a coffee, and doing a check in your fridge to see what you need to eat are all small, everyday changes that people can make to improve themselves. It's important to recognize that these changes compound. They all make us better people and turn us from passive consumers of the economy into active producers of the future. Your actions are contagious. When your friends and family see you doing them, they will often follow your lead. You can silently shift the entire culture by doing the things that are right for the world, and when enough people are on board, the markets will have to change with you. Don't be waiting for a global government to make a huge change for the better. The power, the sheer power, to reduce carbon emissions by billions of tons is in your hands. It's always been in your hands. It's time to make some changes.